Product Proposal

TODO:
Work on determining who the target market is, and who we are NOT targeting. We need to work on defining product market fit to make sure we target the correct customers.

Cyphera Pay

Product & Technical Proposal

Cyphera Corp (Delaware C-Corp)

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1. Executive Summary

Cyphera Pay is an enterprise-grade subscription management platform that brings **Web3 recurring payments** to traditional businesses. Built by **Cyphera Corp**, a Delaware C-Corp, the platform uses:

1. **ActaLink** for account abstraction and Smart Contract Account deployments.
2. **Request Finance** to seamlessly convert stablecoin payments into fiat currency.
3. **Programmable smart contracts** that mirror typical Web2 subscription flows yet remain decentralized and trustless.

The **core idea** is to enable companies like Netflix, Spotify, or SaaS vendors to **offer crypto-based subscriptions** (e.g., USDC payments) without needing to hold crypto themselves—**Cyphera Pay** automatically routes stablecoins through **Request Finance**, converting them to fiat and depositing the proceeds directly into the vendor’s bank account. The platform thus combines **best-in-class Web3 infrastructure** with a **Stripe-like** user and admin experience.

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2. Product Overview

2.1 Market Need

Despite growing interest in cryptocurrency adoption, most Web2 subscription businesses face significant challenges integrating crypto-based payments:

- **Regulatory Complexity**: The burden of compliance and volatility in digital assets.
- **Technical Overhead**: Complexity in handling on-chain transactions, recurring billing, and stablecoin conversion.
- **UX Gap**: Traditional recurring payments rely on frictionless monthly or annual charges, while most crypto payment flows require repeated wallet confirmations.

Cyphera Pay addresses these points by giving businesses a **familiar subscription management experience** while automating all the Web3 intricacies—**account abstraction**, **meta-transactions**, and **fiat settlement**—on the backend.

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3. Architectural Overview

3.1 System Diagram

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+-------------------------+          +-------------------------+
|       Web2 Vendor       | <-----> |     Cyphera Pay API     | <-----> ActaLink (SCA, Paymaster, Relayer)
| (e.g., Netflix, Spotify)|          | & Backend Services      |               |
+-------------------------+          +-------------------------+               |
          |                          |                                  +----------------------+
          |                          v                                  | Request Finance (Fiat)
          |                 +--------------------------+                 +----------------------+
          |                 | Cyphera Pay Dashboard    |
          |                 +--------------------------+
          |
          v
  +-------------------------+
  | End-User Wallet (EOA)   |
  +-------------------------+
```

3.2 Major Components

1. **Cyphera Pay Backend**
    - **Subscription Lifecycle**: Create, update, or cancel subscription plans.
    - **Analytics & Logs**: Track transactions, logs, invoices, and user details.
    - **Integration Layer**: Communicates with **ActaLink** for contract deployment, handles stablecoin invoicing via **Request Finance**.

2. **ActaLink (Smart Contract Account & Account Abstraction)**
    - Deploys **Smart Contract Accounts (SCAs)** for each subscription configuration.
    - Uses **Paymasters** and **Relayers** to sponsor and bundle transactions, eliminating repeated end-user approvals or gas fees.
    - Provides advanced features like social recovery and session keys (if configured).
3. **Request Finance**
    - **Stablecoin Invoicing**: Automates invoice creation and real-time stablecoin-to-fiat conversion.
    - **Vendor Onboarding**: Vendors set up their bank account details and KYC requirements directly with Request Finance.
    - **Payment & Settlement**: Receives stablecoin payments, converts them, then remits fiat to vendors' bank accounts.
4. **Vendor-Facing Dashboard**
    - **Stripe-Like UI** for subscription and user management (e.g., plan tiers, usage analytics, invoices, user activity).
    - **API Key Management** for integration with vendor systems (like Netflix or SaaS apps).
5. **End-User Portal** (Optional in MVP)
    - **User Subscription Management**: View active subscriptions, sign new agreements, or cancel existing plans.
    - Not strictly required for MVP if the vendor has their own front-end, but beneficial for a frictionless experience.

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4. Core Platform Functionality

4.1 Smart Contract Account Creation & Management

- **One-Time Authorization**:
  Users sign a transaction once to subscribe. ActaLink's
  account abstraction logic ensures recurring payments are processed automatically on-chain.
- **Plan Upgrades & Downgrades**:
  If a user wants to change subscription tiers, a new on-chain signature and new Smart Contract Account deployment are triggered. The older contract automatically becomes inactive or is set to expire.

4.2 Automatic Recurring Billing

- **Recurring Charges**:
  Billing intervals (daily, monthly, yearly) are programmed into the subscription SCAs via
  ActaLink.
- **Paymaster Services**:
  Gas fees can be sponsored by a paymaster, preventing wallet prompts for each cycle.

4.3 Stablecoin-to-Fiat Conversion (Request Finance)

- **Stablecoin Payment**:
  Users pay with stablecoins such as
  USDC. These funds go through **Request Finance** for conversion.
- **Real-Time Settlement**:
  Request Finance automatically converts stablecoins to the vendor's chosen fiat (USD, EUR, GBP, etc.) and transfers it to their linked bank account.
- **Invoice Generation**:
  Request Finance can automatically generate an invoice each billing cycle, storing transaction data for reconciliation or compliance.

4.4 Dashboard & API

- **Plan Creation**:
  Vendors define subscription tiers, pricing intervals, and optional add-ons.

- **Vendor Workspace**:
  Each vendor can manage multiple applications or services, each with distinct subscription offerings.
- **Analytics & Reporting**:
  Track MRR, churn, active subscribers, invoice statuses, and other real-time metrics through a vendor-centric dashboard.
- **User Management**:
  Log user wallet addresses (EOAs), optional personal data (name, email), and subscription status.

4.5 KYC & Compliance

- **Vendor-Configurable**:
  Vendors decide how much user data they collect (name, address, etc.) based on regulatory needs.
- **Request Finance**:
  Handles fiat conversion as a licensed money service business. Both vendors and Cyphera Pay can leverage Request Finance's existing KYC and AML compliance protocols.

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5. Implementation Roadmap

5.1 Phase 1: MVP (Est. 3-4 Months)

1. **Cyphera Pay Backend & API**
   - Implement initial subscription management logic.
   - Develop REST/GraphQL endpoints for vendor operations (create/update plans, manage users).
2. **ActaLink Integration**
   - Integrate with ActaLink for SCA deployment, paymaster configuration, and relayer logic.
   - Test user subscription flows on a testnet (e.g., Goerli, Polygon’s test networks).

3. **Request Finance Integration**
    - Leverage stablecoin invoicing for recurring subscriptions.
    - Automate conversion of stablecoins (USDC) to fiat.
    - Provide vendors with an onboarding path to link bank routing details.
4. **Dashboard MVP**
    - Core vendor-facing UI to create subscription plans and view active subscribers.
    - Basic analytics (total subscriptions, MRR, invoice logs).
5. **Invoicing & Logging**
    - Automated generation of invoices from Request Finance each cycle.
    - Store logs in a database, accessible via the dashboard.

6. KYC Integrations

- Provide optional plug-ins for external identity verification.
- Support scenarios where vendors require advanced user identity checks.

5.2 Phase 2: Beta & Enhanced Feature Set (Est. 2-3 Months After MVP)

1. **End-User Portal**
    - Build a user-facing dashboard or embed a widget for frictionless subscription handling.
    - Enable wallet-based SSO with account abstraction features like session keys.
2. **Advanced Analytics & Reporting**
    - Detailed usage charts, churn rates, cohort analysis.
    - Data export (CSV, PDF).
3. **Notification System**
    - Email-based notifications for invoice receipts, subscription changes.

- Possible integration with **XMTP** for direct on-chain notifications to the user's wallet address.

5.3 Phase 3: Expansion & Partnerships

- **Stripe Integration**:
  Hybrid approach allowing simultaneous crypto and credit card billing for vendors already on Stripe.
- **Multi-Chain Deployment**:
  Expand beyond Ethereum mainnet to L2 or EVM-compatible chains (Polygon, Arbitrum, Optimism).
- **Advanced Programmatic Billing**:
  Usage-based billing (per API call, per data usage) or micropayments.
- **Acquisition or Partnership**:
  Position for a potential buyout by Stripe, PayPal, or other leading payment solutions.

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6. Monetization & Pricing Strategy

1. **Transaction Fee**:
   - A small percentage of every transaction processed, deducted automatically before fiat settlement.
2. **Vendor Subscription Tiers**:
   - Base monthly platform fee scaling with usage (active subscriptions, number of apps).
   - Premium tiers offering white-label dashboards, advanced analytics, and priority support.
3. **Add-On Features**:
   - Enhanced KYC/AML modules, advanced reporting, integration with specialized DeFi protocols.

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7. Competitive Advantage

- **ActaLink-Driven Account Abstraction**: Minimizes friction for recurring billing with sponsored gas fees and meta-transactions.
- **Real-Time Fiat Conversion via Request Finance**: Vendors avoid direct crypto custody and volatility.
- **Developer-Friendly**: Well-documented APIs, a robust admin dashboard, and potential off-the-shelf integration for quick adoption.
- **Non-Custodial**: Cyphera Pay does not hold user funds, simplifying regulatory concerns.

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8. Detailed Conclusion

Cyphera Pay stands out as an end-to-end solution that merges **Web3 innovation** with **Web2 convenience**. Here's the overarching value proposition:

- **For Businesses**:
  1. **Immediate Fiat Settlement**: Zero headaches over crypto volatility. Vendors receive predictable, stable fiat funds.
  2. **Subscription Management**: Full-featured dashboard similar to Stripe, but with Web3 under the hood.
  3. **Simplicity & Scalability**: Plug-and-play API integration means less overhead—no direct on-chain management required.
- **For End-Users**:
  1. **Seamless Onboarding**: A single signature starts their subscription, no monthly prompts.
  2. **Choice of Payment Method**: Pay with stablecoins (like USDC) while the vendor receives fiat.
  3. **Control & Transparency**: EOA-based approach ensures users always maintain ownership and see on-chain logs.
- **For Investors & Partners**:
  1. **Big Market Potential**: Bridging crypto and subscription economy unlocks new revenue streams for Web2 and Web3 adopters.

2. **Strategic Position**: A potential acquisition target by top payment processors (Stripe, PayPal, etc.).
3. **Flexible Expansion Path**: Expand to multiple blockchains and integrate advanced features like usage-based billing, social recovery wallets, NFT-based memberships, etc.

In short, **Cyphera Pay** aspires to become the **go-to** recurring payments gateway for both Web2 and Web3 enterprises. By embedding **ActaLink**'s advanced account abstraction features and leveraging **Request Finance** for smooth stablecoin-to-fiat conversion, Cyphera Pay forms a unique ecosystem that addresses the **core pain points** of crypto subscriptions. This framework drives our roadmap, ensuring robust functionality at launch and ample room for innovation as blockchain technology and global payment standards evolve.

Term Definitions / Dictionary Reference

Account Abstraction: A concept in Web3 where smart contract accounts can handle logic that typically resides in externally owned accounts (EOAs). This allows for features like meta-transactions, gas sponsorship, and custom authorization flows without repeatedly asking for user signatures.

ActaLink: A developer platform that provides tools and APIs for account abstraction. ActaLink handles Smart Contract Accounts (SCAs), paymasters, relayers, and other advanced on-chain features, streamlining decentralized application development.

API (Application Programming Interface): A set of functions and protocols enabling communication between software systems. Vendors use Cyphera Pay's API to create subscription plans, manage users, and interact with on-chain data without directly accessing smart contracts.

Batched Transactions: The process of grouping multiple on-chain operations into a single transaction. This reduces gas fees and the number of signature requests from users.

Custodial / Non-Custodial:

- **Custodial** refers to an entity (like an exchange) that holds and manages users' crypto assets.

- **Non-Custodial** means users control their own private keys, with the service provider never holding assets directly.

Dashboard: A web-based interface providing administrators (vendors) with access to subscription management features, plan settings, usage analytics, invoices, and user management tools.

EOA (Externally Owned Account): A standard cryptocurrency wallet address controlled by a user's private key (e.g., MetaMask). Unlike smart contract accounts, EOAs do not contain programmable logic.

Fiat: Government-issued currency such as USD, EUR, or GBP. "Fiat settlement" refers to converting crypto into traditional currency and depositing it into a linked bank account.

KYC (Know Your Customer): A regulatory requirement for verifying customers' identities. This may involve collecting personal data like name, address, and date of birth.

Meta-Transaction: A technique that allows users to initiate blockchain transactions without paying gas directly. A separate entity (relayer or paymaster) broadcasts user-signed transactions to the network on their behalf.

MRR (Monthly Recurring Revenue): A metric in subscription management indicating the predictable, recurring monthly revenue generated by active subscribers.

Paymaster: A specialized smart contract or service in account abstraction that sponsors gas fees on behalf of users. This improves user experience by eliminating the need for users to hold native chain tokens to pay gas fees every billing cycle.

Recurring Payment: An automated billing model where funds are debited on a predefined schedule (e.g., monthly or annually) once a user authorizes the initial transaction.

Request Finance: A third-party platform that automates the conversion of stablecoins (e.g., USDC) to fiat. Used by Cyphera Pay to settle vendors' crypto payments in their bank accounts.

Relayer: A service or smart contract that bundles and broadcasts user-signed meta-transactions. In the context of subscription billing, a relayer ensures that recurring payments can go through on-chain without requiring repeated user approvals.

Session Keys: Temporary cryptographic keys that grant limited permissions for certain blockchain operations. They help streamline user flows by allowing changes to subscriptions (e.g., upgrades) without requiring the main private key each time.

Smart Contract Account (SCA): An on-chain account controlled by programmable logic rather than a private key. SCAs enable advanced features like automated subscription billing, custom permissions, and paymaster integration.

Stablecoin (e.g., USDC): A cryptocurrency pegged to a stable asset (usually the U.S. Dollar). Stablecoins reduce price volatility, making them suitable for recurring payments and easier integration with traditional finance systems.

Stripe: A widely used Web2 payment processor known for seamless credit card transactions. Cyphera Pay replicates a similar subscription billing experience in Web3.

XMTP (Extensible Message Transport Protocol): A decentralized messaging protocol that enables secure communication directly between blockchain addresses. Cyphera Pay can optionally use XMTP to send on-chain notifications or invoices to user wallets.

Contact:

Cyphera Corp

Email/Website Placeholder

Prepared: [Date]

This comprehensive proposal blends the functionality of ActaLink for account abstraction and Request Finance for real-time stablecoin-to-fiat conversion, ensuring an end-to-end subscription solution that matches modern Web2 billing experiences.

https://hedgey.finance/

https://sablier.com/

downide is that you need to put upfront colateral, aaccount abstrction is easy

https://mee.biconomy.io/ - account abstraction

> https://www.pimlico.io/

it depends on user distribution - we end up building a core feature for their website

most crypto team is looking at 10-30mil valuation

https://gnosispay.com/

https://fortune.com/crypto/2024/10/22/stripe-announces-1-1-billion-acquisition-of-stablecoin-start-up-bridge/

https://www.youtube.com/watch?v=x9iAD3GZyfM